What is no-code?

No-code is a software development approach that enables users to create applications and automate business processes without writing code.

It uses visual interfaces and drag-and-drop tools that enable non-technical users to build functional software with little to no programming knowledge.

The power of no-code development lies in its ability to democratize software creation and empower line of business users to quickly develop solutions that are tailored to their specific needs. For example, consider a marketing team that is looking to build a customer survey app. Using a no-code platform, they can assemble the app in hours instead of weeks by dragging and dropping modules such as question fields, buttons and data collection tools onto a visual canvas.

No-code platforms use drag-and-drop interfaces, visual workflows and pre-built components to enable non-technical users (also known as citizen developers or business technologists) to design, build, test and launch custom solutions without involving IT.

This rapid application development approach accelerates the prototyping and deployment of applications, enabling companies to test ideas and implement solutions at unprecedented speeds. It promotes quick responses to market needs, reduces development costs and the burden on IT teams and frees up technical resources for more complex projects. As a result, businesses can innovate faster and adapt more efficiently to changing demands.

No-code development reshapes the technology landscape by putting development tools in the hands of business users with limited technical skills. Industry analyst Gartner forecasts that 70% of new applications will be developed with low-code/no-code technologies by 2025, up from less than 25% in 2020.

This shift fosters a culture of innovation across all organizational levels, from marketing to operations. As companies face mounting pressure to digitize workflows and automate processes, no-code platforms offer a scalable, cost-effective solution. These tools help business leaders meet their digital transformation goals without the need for extensive coding expertise or additional technical resources.

Brief history of no-code platforms

No-code development traces its origins to the visual programming tools and fourth-generation programming languages (4GLs) of the 1980s and 1990s. However, the modern no-code movement truly gained traction in the late 2000s, propelled by advancements in cloud computing and UI design.

The early 2010s saw the introduction of visual app builders, which laid the foundation for today's no-code landscape. As the decade progressed, a surge of specialized no-code tools emerged, catering to diverse needs from creating websites to business process automation. The mid-2010s brought significant improvements in drag-and-drop interfaces and pre-built components, making these platforms increasingly accessible to non-technical users.

The growth in no-code adoption has accelerated the evolution of these platforms, making them more powerful and versatile than ever before. As a result, businesses can innovate faster and more efficiently, driving digital transformation across various sectors without relying solely on traditional coding expertise.

How no-code development works

No-code platforms revolutionize software development by abstracting away coding languages, logic and syntax and enabling non-technical users to build applications through intuitive drag-and-drop interfaces. “No-code” is a bit of a misnomer in that the interfaces and components users interact with on no-code platforms are built with lines of code.

The difference is that the user doesn’t have to write the code—it’s prebuilt into the platform. Instead, users design applications by arranging visual components such as forms, lists and buttons onto a canvas. These actions generate the necessary code automatically, handling data storage, retrieval and processing behind the scenes.

This abstraction allows users to focus on functionality and design while the platform converts their configurations into functional code. Data handling is streamlined with structured data models and pre-built connectors, facilitating seamless integration with external systems and APIs.

Deployment is simplified with one-click options, enabling applications to be launched quickly without complex setups. The platforms manage server provisioning, scaling and updates, keeping applications secure and up-to-date.

While no-code platforms continue to evolve and offer more advanced features, their primary strength lies in enabling non-technical users to build relatively simple applications and workflows without coding.

No-code tools supplement traditional software development rather than replace it. While some can handle more complex tasks, traditional development or low-code solutions are often superior and necessary for more sophisticated solutions.